CS105.3
Database Management Systems

Normalization

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Normalization

The objective of normalization:

"to create relations where every dependency is on the key, the whole key, and nothing but the key".

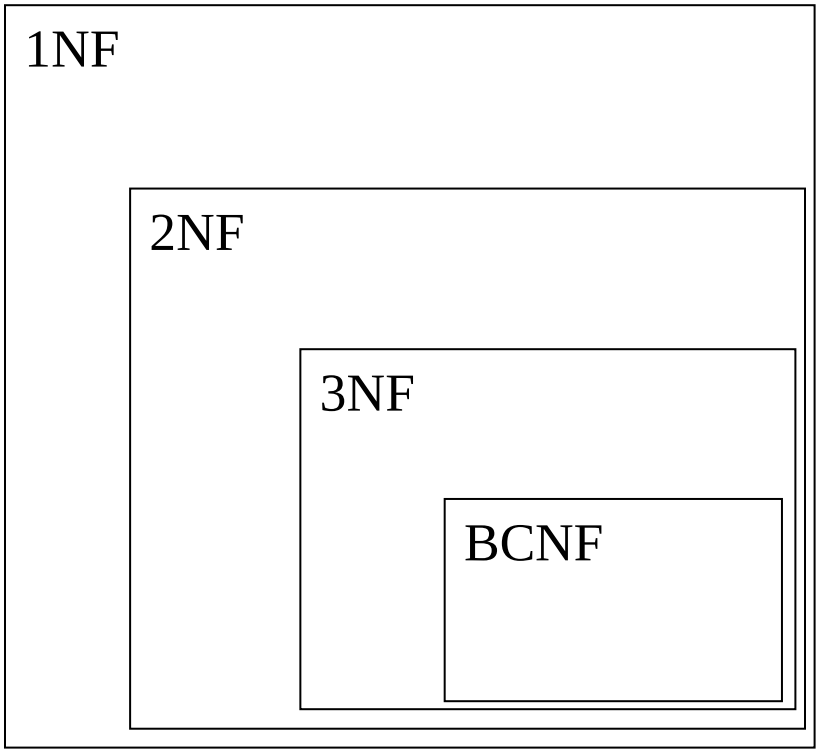
- Evaluating and correcting table structures to minimize data redundancies
- Reduces data anomalies

Normalization

There is a sequence to normal forms:
- 1NF is considered the weakest,
- 2NF is stronger than 1NF,
- 3NF is stronger than 2NF, and
- BCNF is considered the strongest

Also,
- any relation that is in BCNF, is in 3NF;
- any relation in 3NF is in 2NF; and
- any relation in 2NF is in 1NF.

Normalization



a relation in BCNF, is also in 3NF

a relation in 3NF is also in 2NF

a relation in 2NF is also in 1NF

First Normal Form

The following in **not** in 1NF

EmpNum	**EmpPhone**	**EmpDegrees**
123	233-9876	
333	233-1231	BA, BSc, PhD
679	233-1231	BSc, MSc

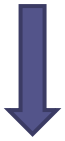
EmpDegrees is a multi-valued field:

employee 679 has two degrees: *BSc* and *MSc*

employee 333 has three degrees: *BA*, *BSc*, *PhD*

First Normal Form

EmpNum	EmpPhone	EmpDegrees
123	233-9876	
333	233-1231	BA, BSc, PhD
679	233-1231	BSc, MSc



1NF

EmpNum	EmpPhone	EmpDegree
123	233-9876	
333	233-1232	BA
333	233-1232	BSc
333	233-1232	PhD
679	233-1231	BSc
679	233-1231	MSc

Exercise

- Convert below table in to 1NF

Emp_ID	Emp_Name	Qualifications	Extension
101	Ann	BSc, MSc	1226
102	Tom	BSc	1269
103	Emmy	BSc, MSc, PhD	1358
104	James	BSc	1457

Second Normal Form

- For a table to be in the Second Normal Form,
  - It should be in the First Normal form.
  - And, it should not have **Partial Dependency**.

Dependency

- Determination
  - “A determines B” indicates that if you know the value of attribute A, you can look up (determine) the value of attribute B.
    - A→B

- Functional Dependency:
  - The attribute B is functionally dependent on the attribute A if each value in column A determines one and only one value in column B.

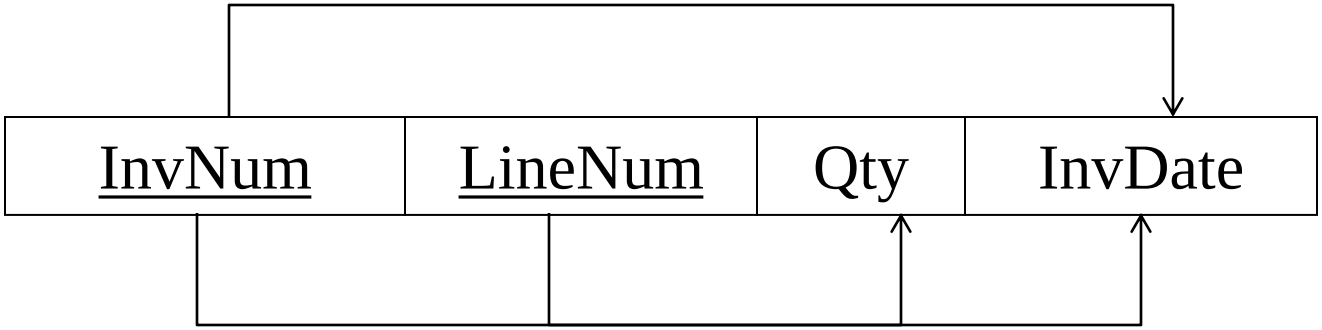
subject_id	subject_name	teacher
1	Java	Java Teacher
2	C++	C++ Teacher
3	Php	Php Teacher

Dependency contd.

- Fully functional dependency (composite key)
  - If attribute B is functionally dependent on a composite key A but not on any Subset of that composite key, the attribute B is fully functionally dependent on A.

Partial dependency

A **partial dependency** exists when an attribute B is functionally dependent on an attribute A, and A is a component of a multipart candidate key.



Candidate keys: {InvNum, LineNum} InvDate is *partially dependent* on {InvNum, LineNum} as InvNum is a determinant of InvDate and InvNum is part of a candidate key

Sample employee working table for identifying partial dependency

Project No	Employee No	Employee Name	Employee Project Work Hours
11	110	Michele	10
12	120	Tina	18
13	110	Michele	15
14	115	Gerard	12
15	121	Neil	14

Exercise

- Determine whether there are any partial dependencies in the below table which is in 1NF and convert it into 2NF.

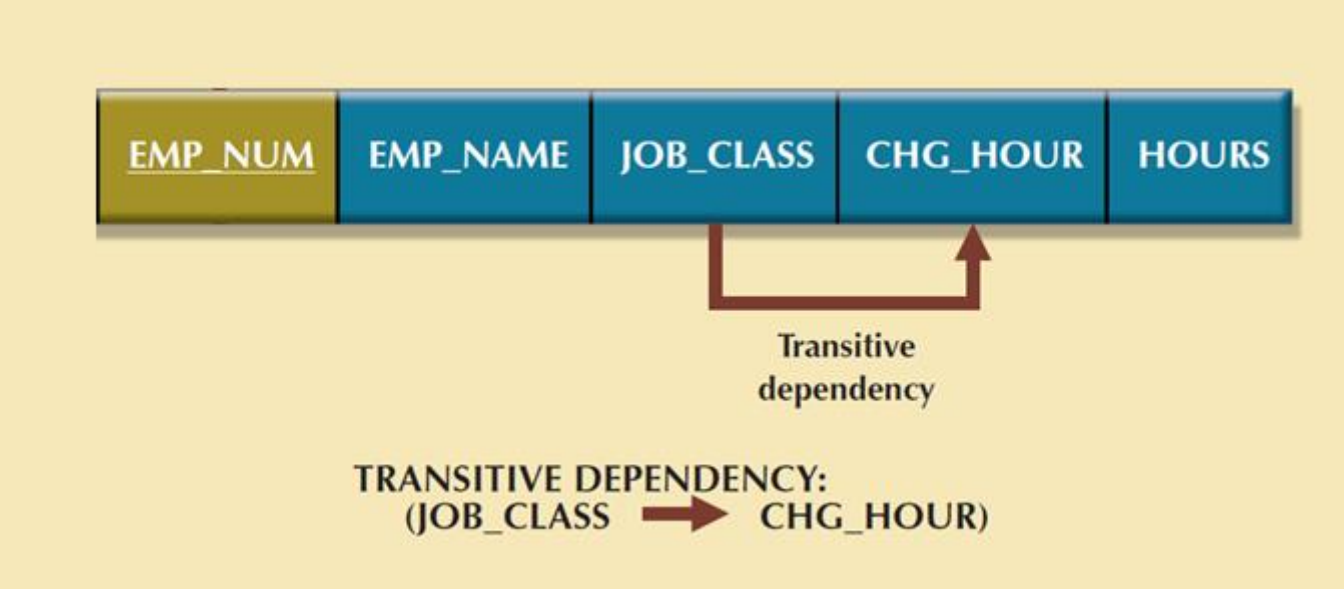
Student_ID	Subject_ID	Subject_Name	Marks	Teacher
10	1	Java	70	Java teacher
10	2	C++	75	C++ teacher
11	1	Java	80	Java teacher
12	3	PHP	85	PHP teacher

Conversion to Third Normal Form

- Steps
  - Make new tables to eliminate transitive dependencies
    - **Determinant**: Any attribute whose value determines other values within a row
  - Reassign corresponding dependent attributes
- Table is in 3NF when it:
  - Is in 2NF
  - Contains no **transitive dependencies**

Transitive dependency

Occurs when a non-prime attribute depends on other non-prime attributes rather than depending upon the prime attributes or primary key.



Exercise

- Determine whether there are any transitive dependencies in the below table which is in 2NF and convert it into 3NF.

UnitCode	UnitName	CourseCode	CourseName
101	Database	COMP2009	Computing
131	Network	COMP2009	Computing
112	OS	COMP2009	Computing
123	Project	BIS3001	Business Management

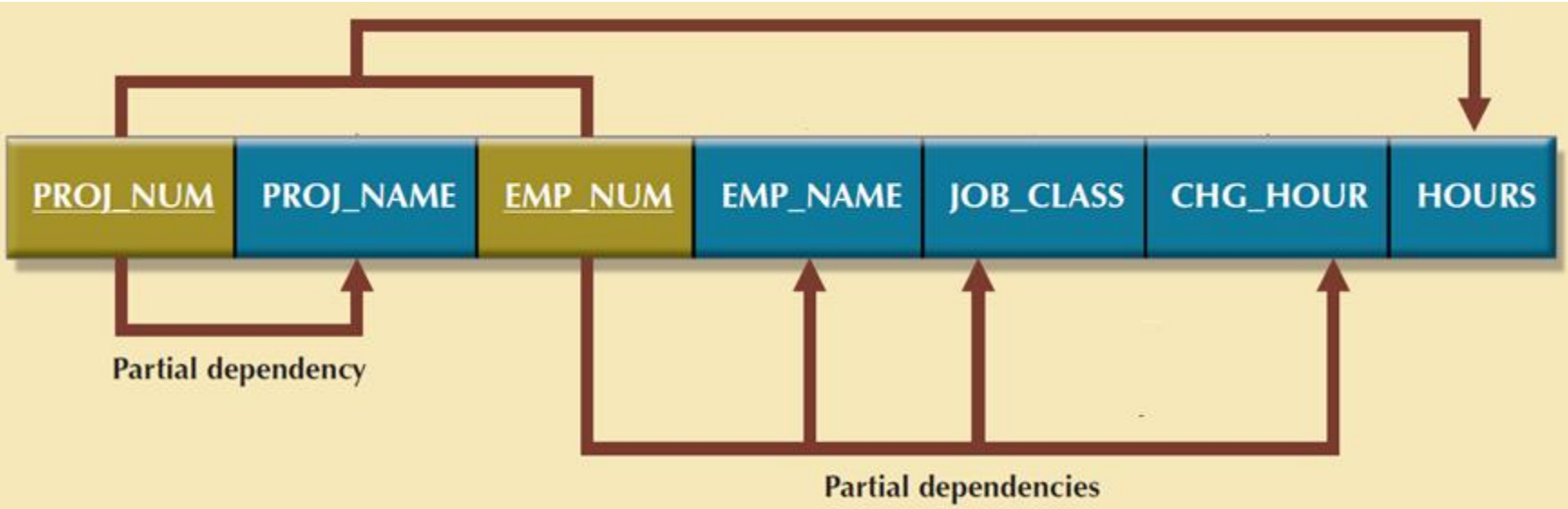
Example – Let's normalize below relation

- The table given is in the 1NF.

Table name: DATA_ORG_1NF **Database name: Ch06_ConstructCo**

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	96.75	48.4
22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6
22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	26.87	22.0
22	Rolling Tide	106	William Smithfield	Programmer	35.75	12.8
25	Starflight	107	Maria D. Alonzo	Programmer	35.75	24.6
25	Starflight	115	Travis B. Bawangi	Systems Analyst	96.75	45.8
25	Starflight	101	John G. News *	Database Designer	105.00	56.3
25	Starflight	114	Annelise Jones	Applications Designer	48.10	33.1
25	Starflight	108	Ralph B. Washington	Systems Analyst	96.75	23.6
25	Starflight	118	James J. Frommer	General Support	18.36	30.5
25	Starflight	112	Darlene M. Smithson	DSS Analyst	45.95	41.4

First Normal Form (1NF) Dependency Diagram



1NF (PROJ_NUM, EMP_NUM, PROJ_NAME, EMP_NAME, JOB_CLASS, CHG_HOURS, HOURS)

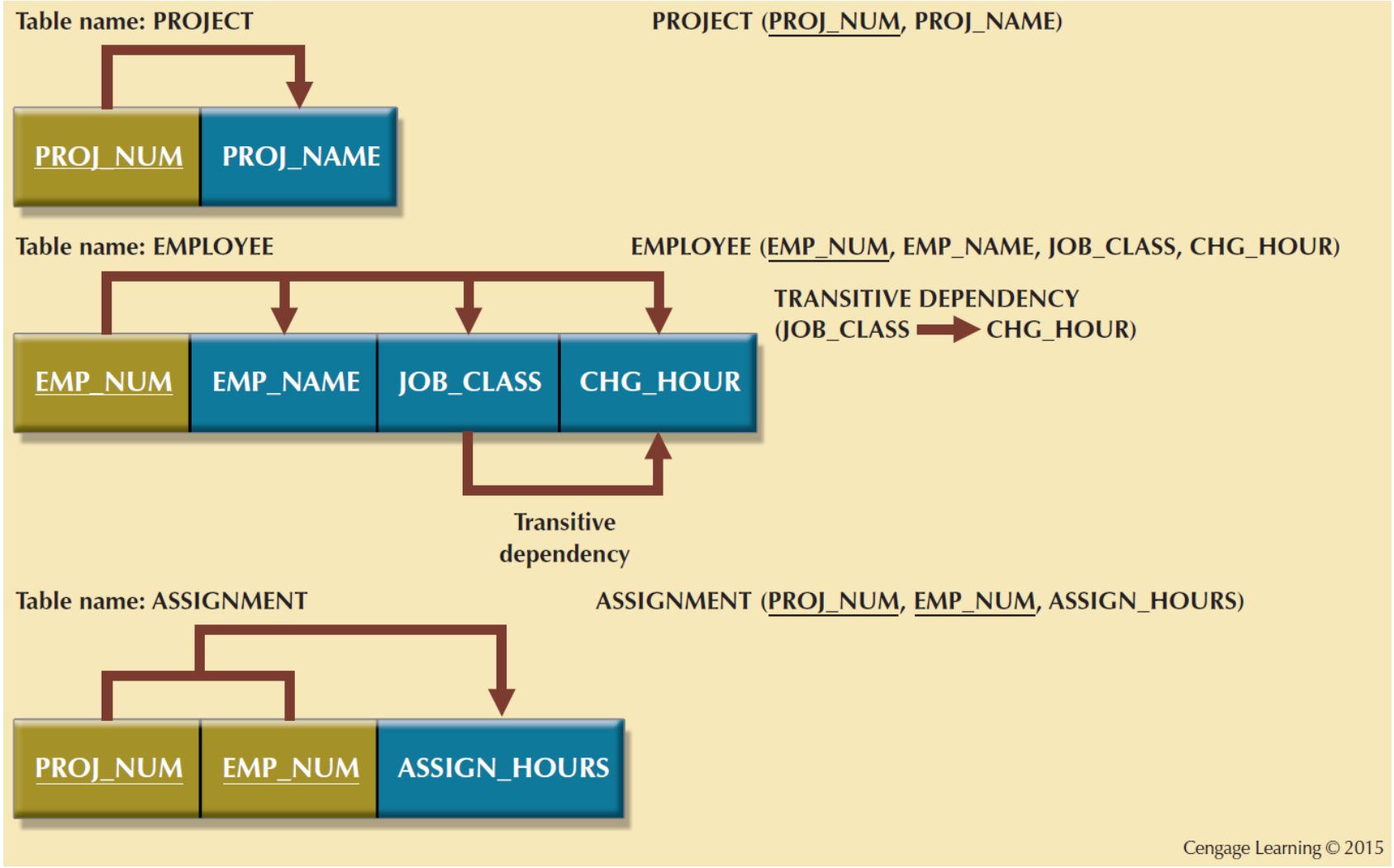
PARTIAL DEPENDENCIES:

(PROJ_NUM → PROJ_NAME)

(EMP_NUM → EMP_NAME, JOB_CLASS, CHG_HOUR)

(EMP_NUM, PROJ_NAME → HOURS)

Second Normal Form (2NF) Conversion Results



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Third Normal Form (3NF) Conversion Results

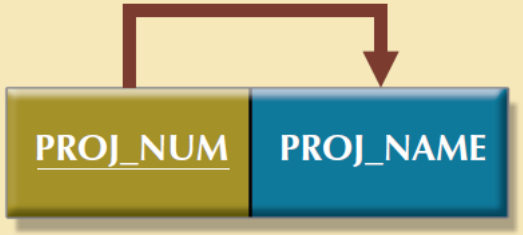


Table name: PROJECT

PROJECT (PROJ_NUM, PROJ_NAME)

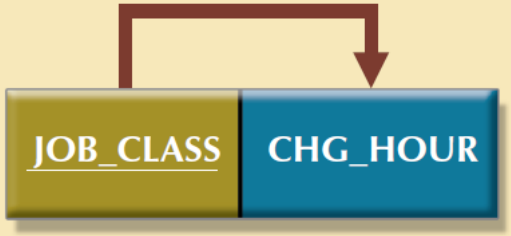


Table name: JOB

JOB (JOB_CLASS, CHG_HOUR)

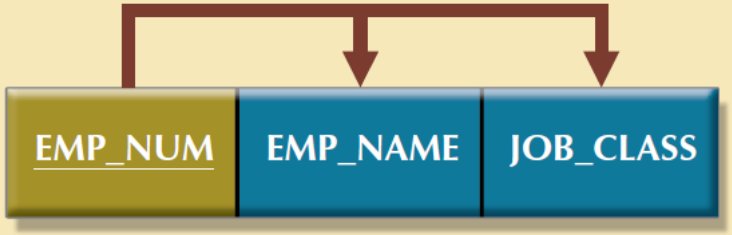


Table name: EMPLOYEE

EMPLOYEE (EMP_NUM, EMP_NAME, JOB_CLASS)

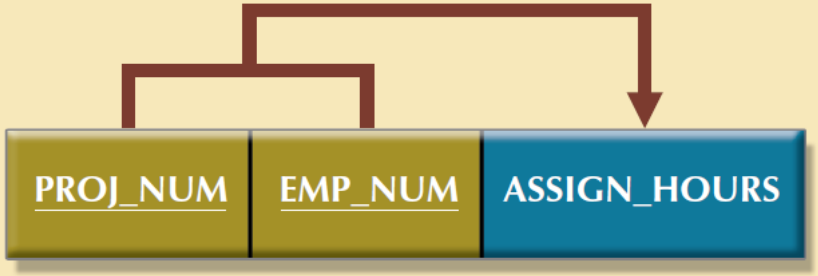


Table name: ASSIGNMENT

ASSIGNMENT (PROJ_NUM, EMP_NUM, ASSIGN_HOURS)

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Identify the 3NF tables for this form

Order Form

Order number: 1234 Date: 11/04/98

Customer number: 9876

Customer name: Billy

Customer address: 456 HighTower Street

City–Country: Hong Kong, China

ProductNo	Desscription	Quantity	Unit Price
A123	Pencil	100	$3.00
B234	Eraser	200	$1.50
C345	Sharpener	5	$8.00

End of the session

SHORT BREAK